

Project Spec

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1 General Description of the Project

1.1 Introduction

Maintaining a healthy lifestyle can be challenging when balancing work, household, and other daily commitments. People often run out of meal ideas or feel uncertain about whether they are making healthy food choices.

RecipeWise is a web application that helps users quickly find healthy recipes tailored to their preferences and needs. By simplifying recipe discovery and meal planning, the platform helps users save time, reduce decision fatigue, and maintain healthier eating habits.

Key features of RecipeWise include a personalized recipe search engine and save for later capability. Users can enter keywords into the search bar to find recipes that match their interests.

Once a search is completed, RecipeWise displays a curated list of recipes that match the user's criteria. Each result contains essential information, such as the recipe name, and brief description, allowing users to compare options.

From the result list, users can select a recipe to view it completely in the recipe detail page, where the recipe name, save for later button, ingredients and step-by-step instructions are located. This experience helps users easily discover, compare, and prepare meals that fit their preferences.

1.2 External Mechanisms

- TheMealDB API (<https://www.themealdb.com/>)
 - Recipe search
 - Recipe Catalog
 - Recipe details
 - Ingredient Information
- Supabase
 - Favorites storage

1.3 GUI

RecipeWise will provide a web-app experience using Flask.

1. Home Page
 - Search bar
2. Search Results Page
 - Search bar
 - Recipe cards list
3. Recipe Details Page
 - Recipe image
 - Ingredients list
 - Step-by-step instructions
 - Favorite button
4. Favorites Page
 - Saved recipes list

2 Task Vignettes

Goal: Luisa wants to quickly find a recipe that uses ingredients she already has at home, save recipes she enjoys, and easily access them in the future using RecipeWise.

Situation: Luisa notices she has several carrots left in her refrigerator and wants to use them before they go to waste. Because she has limited time, she needs a recipe that is both easy and quick to prepare. After finding and cooking a recipe she enjoys, she saves it to her favorites so she can make it again later. One week later, she returns to RecipeWise to locate the saved recipe.

2.1 Task 1: Search a meal

User Goal: Find recipes that use carrots.

1. User opens RecipeWise.
2. User locates the search bar on the home page.
3. User enters "carrots" into the search bar.
4. User clicks the search button.
5. User views the list of recipes containing carrots.
6. User selects a meal.
7. App displays ingredients, instructions, image, and other available details.



Fig 1. Home Page

2.2 Task 2: Save a Meal

User Goal: Save a recipe for future use.

1. User opens a recipe.
2. User reviews the ingredients and instructions.
3. User decides to save the recipe after a positive cooking experience.
4. User clicks the *<Save to Favorites> button*.

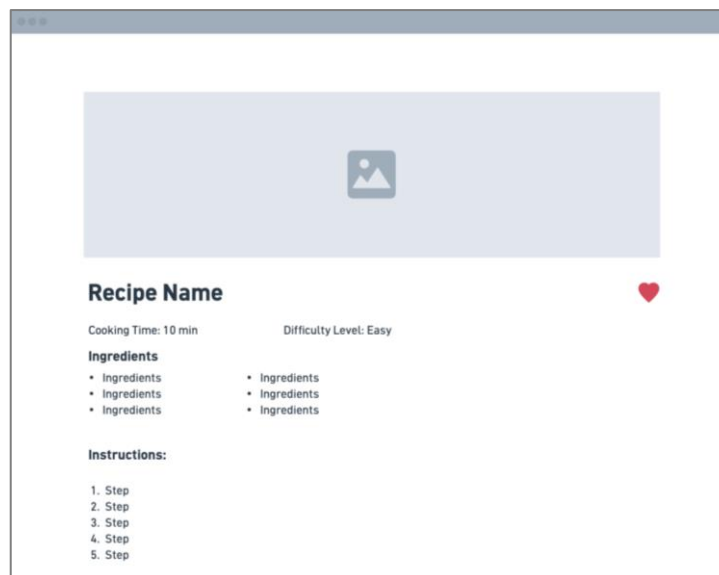


Fig 4. Recipe Details Page – Saved Recipe

2.3 Task 3: See saved meals

User Goal: Find a previously saved recipe.

1. User opens RecipeWise one week later.
2. User navigates to the favorites page.
3. User views the list of saved recipes.
4. User locates the carrot recipe she is looking for.
5. User opens the recipe details page.

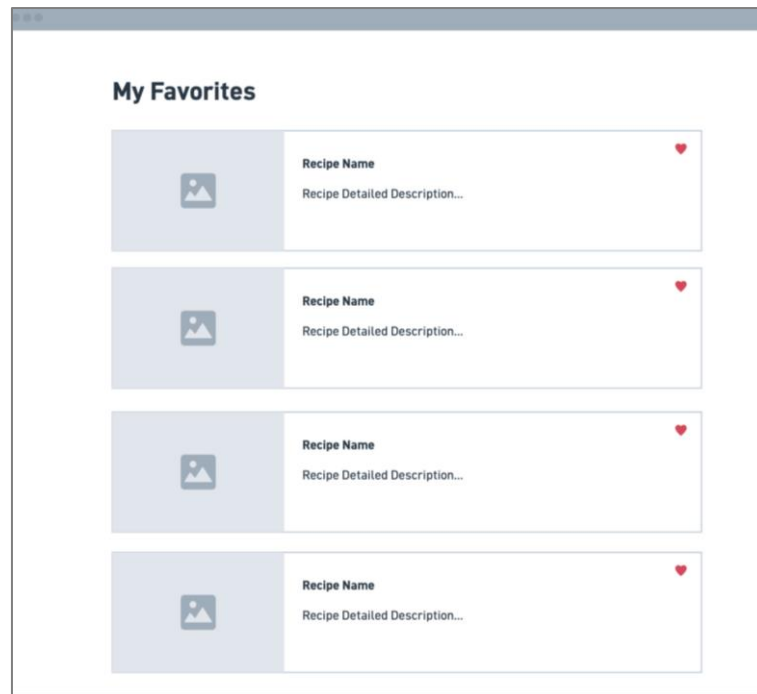


Fig 5. My Favorites Page

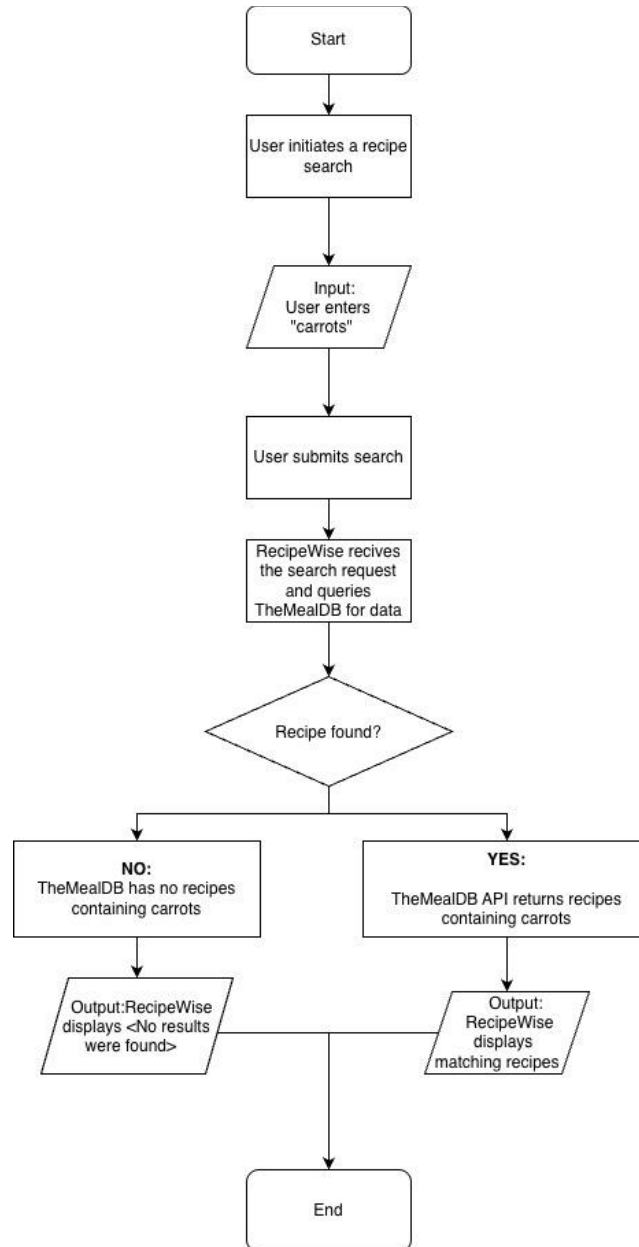
3 Technical Flow

3.1 Project Scope Limitation

User authentication and account management are not included in this project. The application assumes the user is already authenticated, allowing the project to focus on recipe search, recipe details, and recipe persistence.

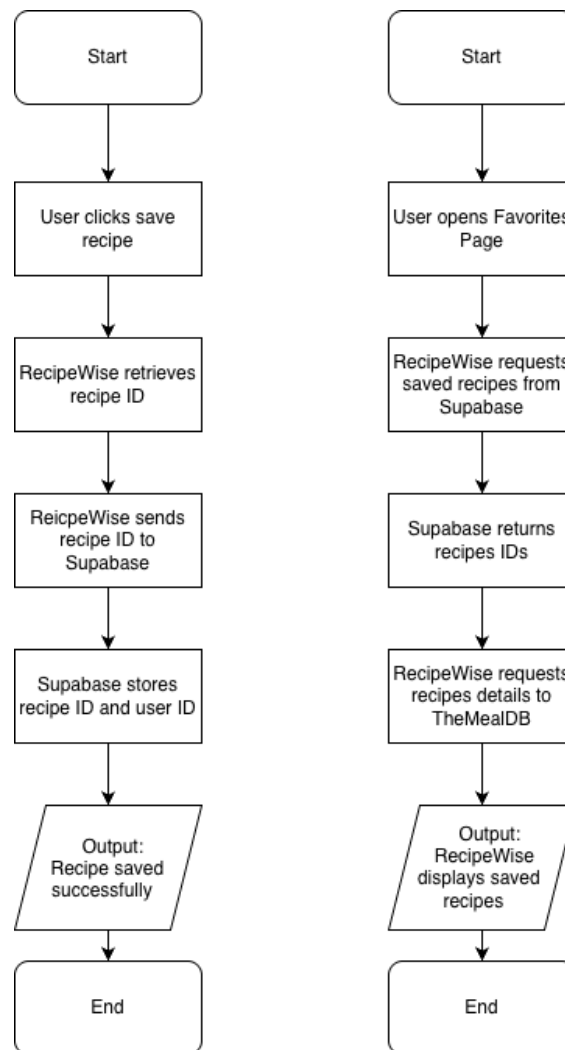
3.2 Search Functionality

- Users enter a search keyword (input string).
- RecipeWise sends the search keyword to TheMealDB API.
- TheMealDB returns recipe data.
- RecipeWise displays the search results to the user (output)



3.3 Save Recipe Functionality

- User selects saved recipe.
- RecipeWise retrieves the user ID and recipe ID.
- RecipeWise sends the information to Supabase.
- Supabase stores information (user ID and recipe ID).
- The recipe ID is used to retrieve the saved recipe list.



4 Self-Assessment

This homework was very helpful in improving my understanding of the overall logic of the program. It gave me a clearer picture of how RecipeWise will interact with external services (TheMealDB and data storage through Supabase).

One area that still concerns me is integrating Supabase with the application. While the overall search and favorite-saving workflow is clear, I anticipate challenges in connecting external services and managing user data correctly.

Overall, I now have a much clearer understanding of the project's architecture and development process. Creating the technical flows helped me break the application into smaller, more manageable components and better understand how data moves between the user, RecipeWise, TheMealDB, and Supabase. While I am still somewhat nervous about the implementation phase, I feel more prepared and confident than I did before completing this assignment.